Experiment No:10
Experiment to understand array function in Data Science

AIM:

To understand array function in Data science.

DESCRIPTION:

Understand array function using Numpy library

CODE:

```
import numpy as np

# Create an array with different input values
array = np.array([10, 20, 30, 40, 50, 60, 70, 80, 90])
print("Original array:")
print(array)

# Square root
print("\nSquare root:")
print(np.sqrt(array))

# Number of dimensions
print("\nDimensions:")
print(array.ndim)

# Reshape into 3x3
new_array = array.reshape(3, 3)
print("\n3x3 array:")
print(new_array)

# Number of dimensions
print("\nDimensions of new_array:")
print(new_array.ndim)

# Convert to 1D
print("\nRavel:")
print(new_array.ravel())

# Reshape again
newm = new_array.reshape(3, 3)
```

```
print("\nnewm:")
print(newm)

# Select row 3, columns 2 and 3
print("\nnewm[2,1:3]:")
print(newm[2, 1:3])

# Select row 2, columns 2 and 3
print("\nnewm[1:2,1:3]:")
print(newm[1:2, 1:3])

# Select first column
print("\nnew_array[0:3,0:1]:")
print(new_array[0:3, 0:1])

# Select first column of first 2 rows
print("\nnew_array[0:2,0:1]:")
print(new_array[0:2, 0:1])

# Select rows 2 and 3
print("\nnew_array[1:3]:")
print(new_array[1:3])

# Change a value
new_array[0, 0] = 100

print("\nAfter changing first value:")
print(new_array)
```

OUTPUT:

```
Original array:
[10 20 30 40 50 60 70 80 90]

Square root:
[3.16227766 4.47213595 5.47722558 6.32455532 7.07106781 7.74596669
 8.36660027 8.94427191 9.48683298]

Dimensions:
1

3x3 array:
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]]

Dimensions of new_array:
2

Ravel:
[10 20 30 40 50 60 70 80 90]

newm:
[[10 20 30]
 [40 50 60]
 [70 80 90]]

newm[2,1:3]:
[80 90]

newm[1:2,1:3]:
[[50 60]]

new_array[0:3,0:1]:
[[10]
 [40]
 [70]]

new_array[0:2,0:1]:
[[10]
 [40]]

new_array[1:3]:
[[40 50 60]
 [70 80 90]]

After changing first value:
[[100  20  30]
 [ 40  50  60]
 [ 70  80  90]]
```

RESULT:

The array operations were successfully performed using numpy including reshaping ,indexing,slicing and raveling.